

European Marine Omics Biodiversity Network (EMO BON)

June 2021, EMO BON metadata

Time is near for our first EMO BON sampling event and for starting the metadata collection!

The collection of rich and rigorous metadata to accompany the data is high priority for EMO BON. Metadata are the key that places the data in a broader context and ensures the correct data FAIRification. To collect the metadata, we have prepared detailed metadata log-sheets that should be completed by the contact people of each EMO BON station. The log-sheets are prepared as online spreadsheet files, in the EMBRC google account. Each file corresponds to one EMO BON observatory. In addition, there are separate files for the water column and the soft substrates samplings for each observatory.

You will receive an invitation by email for accessing the file(s) that corresponds to your observatory station. We ask that you complete the metadata spreadsheets as soon as possible after the sampling event. This will mark that the sampling took place, that the samples are collected and will create the list of samples that belong to your observatory. The metadata spreadsheet file includes 5 sheets:

1) READ_THIS_FIRST

This sheet includes brief instructions on what the other sheets contain. It also asks that the structure of the file is not changed when completing the metadata. Keeping the structure of the file as is, is of great importance. These files are going to be used for uploading the metadata to public databases and if their structure changes, the upload will not be possible. This means that it is not acceptable to rename the columns and to add or remove columns.

2) observatory

EMO BON metadata are separated in categories as described in the EMO BON Handbook. This sheet contains the Observatory Metadata that are all information regarding the observatory station, the sampling sites, their characteristics, and the contact people. This information was provided by you in the registration forms and has been added in the log-sheets by the EMO BON Science Officer. It will not be necessary to complete this sheet, however, we do ask that you check if the information has been inserted correctly. If you wish to change anything in the observatory sheet, please also inform the EMO BON Science Officer Ioulia Santi (ioulia.santi@embrc.eu) of the change.

3) sampling

EMO BON metadata are separated in categories as described in the EMO BON Handbook. This sheet contains the Sampling Metadata which include all the information related to the specific sampling event and the fate of the samples. For example, information on the date and time of sampling, sample handling details and deviations from protocols are included here. This sheet should be completed by



the stations after every sampling event. Each column represents a different metadata field. Each row corresponds to one sample. The field “source_mat_id” is where the sample label should be added. This is the unique identification for each sample collected and will follow the sample to the data generation.

4) measured

EMO BON metadata are separated in categories as described in the EMO BON Handbook. This sheet contains the Measured Metadata which include all the additional variables measured during sampling. Some variables are characterized as mandatory and are to be measured during every sampling event. Observatories may choose to additionally measure any of the optional variables. For each variable field there is a field for the method description. Observatories can select the way the variables are measured; however, they must provide this information in the metadata. It is also important that the same method is followed for each measured variable for all EMO BON samplings. As the results from the analyses may take time to be produced, observatory stations may complete the Measured Metadata later. Although we ask that you complete the mandatory Measured Metadata as soon as possible after a sampling event. Each column represents a different metadata field. Each row corresponds to one sample. The field “source_mat_id” is where the sample label should be added. This is the unique identification for each sample collected and will follow the sample to the data generation.

5) definitions

This sheet contains the definition of each metadata field included in the sheets “observatory”, “sampling” and “measured”. The metadata fields are listed in alphabetical order. For each metadata field, it is given: its full name, a detailed definition, the value and units that should be used, an example, which metadata category it belongs to, if it is mandatory or optional and if it is mandatory in other checklists.

The contact people of the observatories may contact the EMO BON Science Officer any time for questions and/or assistance when filling in the metadata. Samples for which metadata are incomplete or improperly documented will be rejected and will not be further processed.

Contact:

Ioulia Santi

**European Marine Biological
Resource Centre**

Paris Headquarters

4 Place Jussieu,

Tour 46/00, 1er étage, bureau 101

75252 Paris Cedex 05 (FR)

www.embrc.eu

Email: ioulia.santi@embrc.eu

Phone: +30 2 81 03 37 727

Mobile: +30 6 94 51 78 545

European Marine Omics Biodiversity Network (EMO BON)

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Updating EMO BON metadata spreadsheets

In EMO BON, we are actively and continuously working on improving and standardizing the data management system, while at the same time ensuring adherence to the FAIR data principles. As often discussed, data management begins even before the sample collection. We have paid attention to the proper metadata collection from the beginning of EMO BON, and we have created the online spreadsheets to record all the metadata related to the samples.

From the beginning of EMO BON until now, the spreadsheets have changed and improved, and they now represent the core stepping stones for the whole EMO BON data management. You may have already noticed some changes in the EMO BON metadata spreadsheets. With this note we wish to summarize the changes and give you more details. As requested, you may also find a list of some commonly made errors. Please note that these changes are not yet applied in all online spreadsheets, but we are looking into having all spreadsheets fully updated in the coming months.

We believe that EMO BON metadata collection is among the most thorough and well-structured in the whole marine biology research landscape. It provides the basis not only for advancements in the genomic data management, but also for observation-based scientific questions.

We are thankful to all of you for keeping up with us during this procedure, for making the effort to fill the sometimes complex fields, for wanting to learn how to do this and for actively helping us improve the spreadsheets. We are grateful to the people who worked on this, put a great effort to improve the EMO BON metadata spreadsheets and keep working on our data management procedures: Christina Pavloudi (EMBRC-ERIC), Katrina Exter (VLIZ), Marc Portier (VLIZ), Cymon J. Cox (CCMAR), Bruno Louro (CCMAR), Stelios Ninidakis (HCMR), Melina Loulakaki (HCMR), Maria Luisa Chiusano (UNINA), Andrzej Tkacz (CCMAR), Salva Fernández Bejarano (VLIZ), Laurian Van Maldeghem (VLIZ), Melanthia Stavroulaki (HCMR).

As always, we are happy to hear from you (questions, issues, discussions), so feel free to contact us!



Summary of updates on the EMO BOD metadata spreadsheets

1. The spreadsheets were migrated from the EMBRC secretariat account to an account dedicated to EMO BOD. Editing rights have been given to all relevant people. In case you wish to add more people from your observatories as editors, please contact us.
2. During the update process, some redundant sheets were created (Updated definitions, old-sampling). This was to ensure the passing of all information to the updated spreadsheets. Only necessary sheets will remain: READ_THIS_FIRST, observatory, sampling, measured, definitions.
3. We changed the entry format in the replicate column of the sampling sheet for the blank samples. An underscore was added and the new format is "blank_1", "blank_2" etc.
4. There is a new term to record environmental events taking place during sampling and may have affected the samples. We can take this information into account when analysing the data. If you want to add more than one value, use ; as the separator. The values should be from [ENVO](#) and some examples are:
 - o marine algal bloom [ENVO:02500036]
 - o coral bleaching process [ENVO:01000852]
 - o glacial abrasion [ENVO:01000682]
 - o storm [ENVO:01000876]
 - o drought [ENVO:1000745]
 - o rain [ENVO:01001564]
 - o snow [ENVO:01000406]
 - o hail [ENVO:03400011]
 - o oil spill [ENVO:00002061]
 - o increased temperature [PATO:0001305]
 - o environmental pollution [ENVO:02500036]
 - o biofouling [ENVO:06105023]

Spreadsheet: water, sediment, ARMS

Sheet name: sampling

Term (column) name: noteworthy_env_cond

Definition: If there were any noteworthy/special environmental conditions during the sampling event, please note down the most appropriate term from the Environmental Ontology. Some suggestions for terms are: marine algal bloom [ENVO:02500036], coral bleaching process [ENVO:01000852], glacial abrasion [ENVO:01000682], storm [ENVO:01000876], drought [ENVO:1000745], rain [ENVO:01001564], snow [ENVO:01000406], hail [ENVO:03400011], oil spill [ENVO:00002061], increased temperature [PATO:0001305], environmental pollution [ENVO:02500036], biofouling [ENVO:06105023]. Additional appropriate terms may be selected from here: https://ontobee.org/ontology/ENVO?iri=http://purl.obolibrary.org/obo/ENVO_01000203

Who is adding this information: sampling operators

5. There are a few new terms related to data ownership.

Spreadsheet: water, sediment, ARMS



Sheet name: observatory

Term (column) name: data_owner

Definition: the organization responsible for the curation, integrity, publication and use of generated data

Who is adding this information: EMBRC

Term (column) name: data_owner_edmoid

Definition: EDMO id of the data owner organization

Who is adding this information: EMBRC

Term (column) name: data_contact_email

Definition: data owner organization email

Who is adding this information: EMBRC

6. There is a new term to indicate in which batch each sample belongs to. This term is useful for managing the samples and instantly knowing where to find information related to the next steps

Spreadsheet: water, sediment, ARMS

Sheet name: sampling

Term (column) name: seq_batch

Definition: batch of EMO BON samples processed together for DNA extraction, sequencing, and data acquisition. Example: "batch 1", "batch 2"

Who is adding this information: EMBRC

7. Terms related to the ENA submission of the data have been added.

Spreadsheet: water, sediment, ARMS

Sheet name: observatory

Term (column) name: ENA_accession_number_umbrella

Definition: Umbrella study accession number is a unique identifier given to an umbrella study once the sample is registered to ENA (European Nucleotide Archive). It refers to a specific umbrella study and connected to all the projects, samples, sequencing runs and data included.

Who is adding this information: EMBRC

**Term (column) name: ENA_accession_number_project**

Definition: Project accession number is a unique identifier given to a project once the project is registered to ENA (European Nucleotide Archive). It refers to a specific project and connected to all the samples, sequencing runs and data included.

Who is adding this information: EMBRC

Spreadsheet: water, sediment, ARMS

Sheet name: sampling

Term (column) name: ENA_accession_number_sample

Definition: Sample accession number is a unique identifier given to a project once the sample is registered to ENA (European Nucleotide Archive). It refers to a specific sample and connected to all the sequencing runs and data included.

Who is adding this information: EMBRC

8. We found that the priorly used `source_material_id` was not easy to handle in later data management steps since there were cases of non-unique ids. This is why we simplified the `source_material_id` to include ascending numbers starting with the first sample collected (or added in the spreadsheet) in an observatory. The numbering is separate for each observatory. The new format is `EMOBON_[observatory name]_ SampleType[Wa/So/Ha]_iterator (1,2,3 etc)`, for example `EMOBON_BPNS_Wa_1`. This is created via a google equation. There is no need for the sampling operators to fill in this column, as it will be filled by EMBRC. The intention is to lock this column and only allow EMO BON management team to edit it because this ID, once created, must NEVER be changed (so no moving the rows around!). The common sample name that is easily readable will remain in the spreadsheets under term `source_mat_id_orig` and this is the term sampling operators should be filling in. The `source_mat_id_orig` will always be the first column in each sheet to easily indicate which sample each row includes. Therefore, please add the `source_mat_id_orig` for every sample in the sampling sheet according to the EMO BON Handbook and as written on your sample containers.

Spreadsheet: water, sediment, ARMS

Sheet name: sampling, measured

Term (column) name: source_material_id

Definition: Unique identifier to each row/sample: it consists of the terms 1) Project ("EMOBON") 2) Sampling Site ID; that is the Observatory ID (for example "SMN99") supplemented with the sampling site indicator ("Wa" for water column, "So" for soft substrates and "Ha" for hard substrates) and 3) an iterator (1,2,3 etc). This is created via a google equation. To add this identifier to a new row, simply drag the value from the cell above. This column can only be filled by EMBRC.

Who is adding this information: EMBRC

**Term (column) name: source_material_id_orig**

Definition: A unique identifier assigned to a material sample according to EMO BON Handbook as added by the sampling operators and written on the sample containers. This identifier is not quality controlled but remains unchanged. This identifier consists of 5 terms: 1) Project (“EMOBON”) 2) Sampling Site ID; that is the Observatory ID (for example “SMN99”) supplemented with the sampling site indicator (“Wa” for water column, “So” for soft substrates and “Ha” for hard substrates) 3) Sampling Campaign Date formatted as YYMMDD. For example “220315” would be the campaign on 15 March 2022. Especially for Hard substrates, both the deployment and retrieval dates have to be recorded formatted as YYMMDD-YYMMDD 4) Size fraction (Wa: “3um” / “0.2um” / “20um” / “200um”) or organisms collected (So: “micro” for microorganisms, “meio” for meiobenthos and “macro” for macrobenthos) or ARMS fraction (Ha: “SF” for sessile fraction, “MF500” for motile fraction sieved through 0.5 mm, “MF100” for motile fraction sieved through 0.1 mm) 5) Replicate number (1-4). For labelling the negative control, this term will be replaced by the notation “blank”. All terms must be separated by “_”.

Who is adding this information: sampling operators

9. The definitions of size_frac_low and size_frac_up were swapped so that the first is the lower filter value and the second is the upper filter value. This used to be a problem because the MixS terms checklist had them reversed and we wanted to follow the checklist. We raised this with the Genomic Standards Consortium and the definitions will be corrected in the new MixS version. This is already corrected in the water spreadsheets, and you don’t have to do anything for the samples already entered. Definitions are now:
 - a. size_frac_low: size-fraction lower threshold refers to the mesh/pore size used to retain the sample. Materials smaller than the size threshold are excluded from the sample
 - b. size_frac_up: size-fraction upper threshold refers to the mesh/pore size used to pre-filter/pre-sort the sample. Materials larger than the size threshold are excluded from the sample
10. All terms that are flagged as mandatory in sheet measured are coloured yellow.

Frequently made errors

1. Not double-checking the values added. Please always *double-check* after having filled in values. Please pay attention to not:
 - drag-drop values incorrectly (did you really want all numbers to increase by 1 for each row?)
 - copy-pasting from another row but forgetting to change e.g. the date
2. Make sure all mandatory cells are filled. What is required is
 - either write the actual value
 - or write “NA” if that cell will never have a value (e.g. a sensor broke, and no measurement was taken)
 - or write “expected later” or “expected MM-YY” if the value will come later
3. Using > or < in the size_frac_low and size_frac_up columns: the two values between them indicate a min to max range, so there is no need to use the upper or lower limits symbols.
4. Make sure you are using the indicated units for each measurement. The units are given in the definitions sheet, where terms are ordered alphabetically.



5. The following columns were always built to allow list entries, but please pay attention to how the lists are entered, see the example below

Columns including list entries:

biomass and biomass_method
chem_administration
n_alkanes and n_alkanes_method
organism_count and organism_count_method
particle_class
phaeopigments and phaeopigments_method
pigments and pigments_method

Example:

pigments

B-caroten: 0.04;α-caroten 0.005

pigments_method

HPLC (Alliance HPLC System) following the method described by Zapata et al. 2000; HPLC (Alliance HPLC System) following the method described by Zapata et al. 2000

**Please add as many methods as there are pigments. Even if the method is the same, please add it again)*

For the EMO BON management team,

Ioulia Santi, PhD

**Observation, Data and Service
Development Officer**

**European Marine Biological
Resource Centre (EMBRC-ERIC)**

Paris Headquarters

4 Place Jussieu,

Tour 46/00, 1er étage, bureau 101

75252 Paris Cedex 05 (FR)

www.embrc.eu

Email: ioulia.santi@embrc.eu

Phone: +30 2 81 03 37 727

Mobile: +30 6 94 51 78 545

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